SmartElex LoRa Module Ra- 02 SX1278 433Mhz Breakout Board

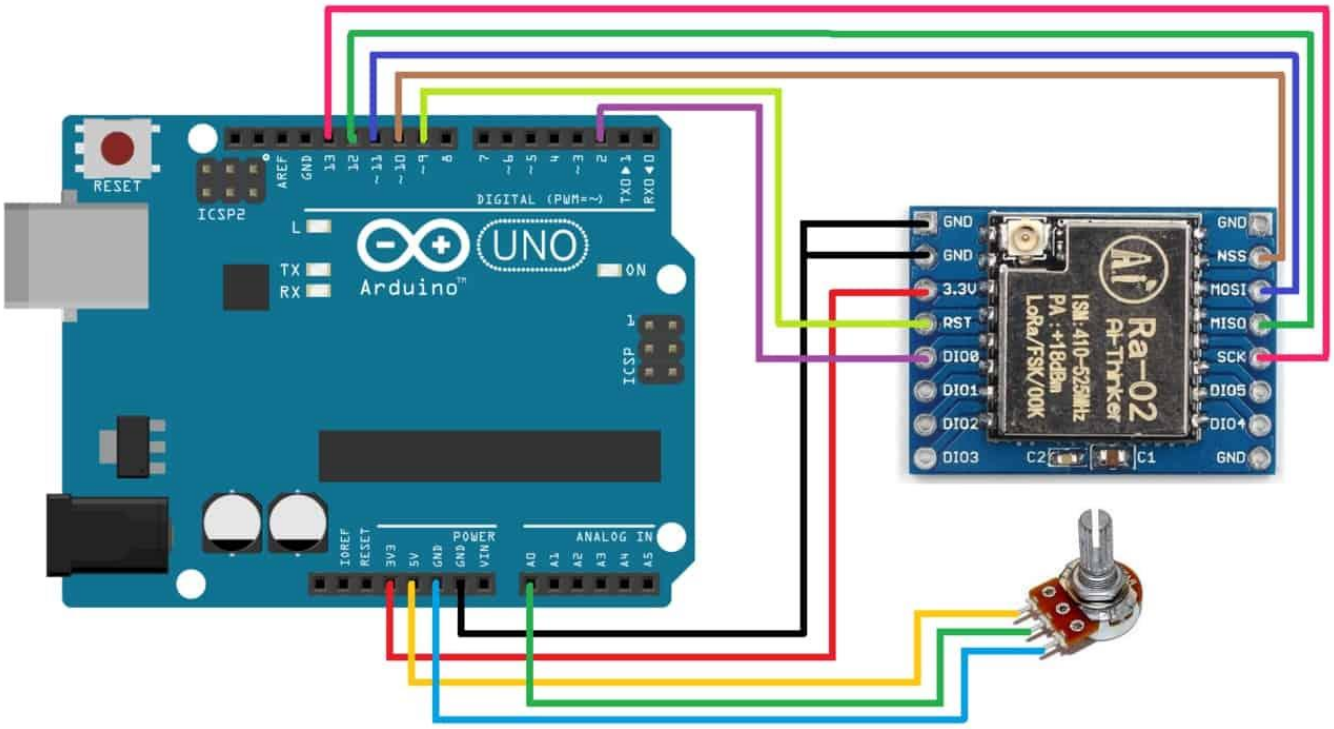
The **SmartElex LoRa Module RA-02 SX1278 433 MHz Breakout Board** is a wireless communication board based on the popular **SX1278 LoRa transceiver**. It operates in the **433 MHz ISM band** and enables long-range, low-power data transmission using LoRa modulation. The breakout design gives easy access to SPI, power, and control pins for rapid integration with microcontrollers. This board is ideal for IoT, telemetry, remote sensing, and other long-distance wireless projects.

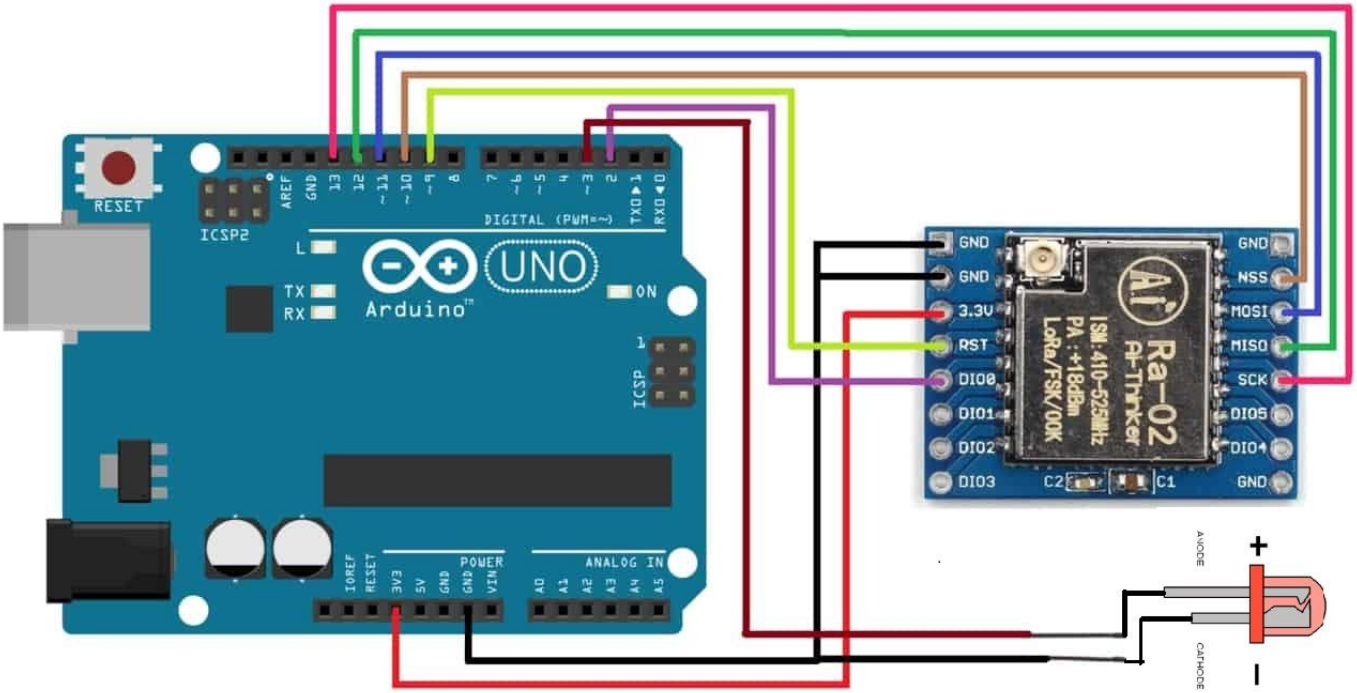
❖ Features

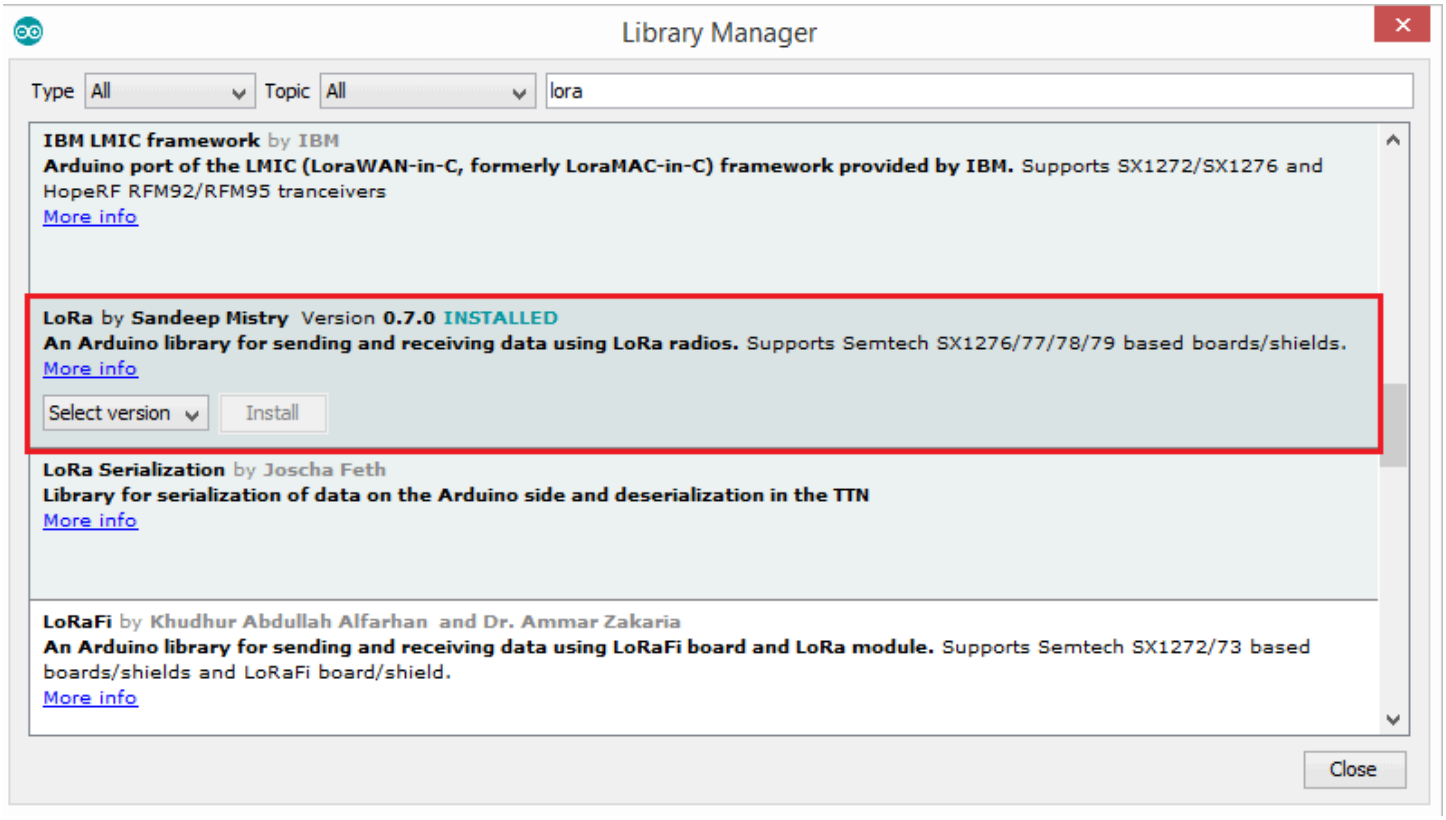
- Provides long-range wireless connectivity with high sensitivity and noise immunity.
- LoRa modulation supports robust communication in noisy environments.
- LoRa ™ Spread Spectrum modulation technology.

❖ Specifications

- Built around the **SX1278 LoRa transceiver** optimized for 433 MHz band communication.
- RF output power typically up to **20 dBm** for extended range.
- Constant RF power output at + 20dBm-100mW voltage change.
- Automatic RF signal detection, CAD mode and very high speed AFC.






Library Manager
Type All
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lora
IBM LMIC framework by IBM
Arduino port of the LMIC (LoraWAN-in-C, formerly LoraMAC-in-C) framework provided by IBM. Supports SX1272/SX1276 and HopeRF RFM92/RFM95 tranceivers
More info
LoRa by Sandeep Mistry Version 0.7.0 INSTALLED
An Arduino library for sending and receiving data using LoRa radios. Supports Semtech SX1276/77/78/79 based boards/shields.
More info
Select version
Install
LoRa Serialization by Joscha Feth
Library for serialization of data on the Arduino side and deserialization in the TTN
More info
LoRaFi by Khudhur Abdullah Alfarhan and Dr. Ammar Zakaria
An Arduino library for sending and receiving data using LoRaFi board and LoRa module. Supports Semtech SX1272/73 based boards/shields and LoRaFi board/shield.
More info
Close

GND
GND
3V3
RST
IO0
IO1
IO2
IO3
GND
NSS
MOSI
MISO
SCK
IO5
IO4
GND
GND
GND
3.3V
RST
IO0
IO1
IO2
IO3
GND
NSS
MOSI
MISO
SCK
IO5
IO4
GND
Ra-02
Ai-Thinker
ISM:410-525MHz
PA:+18dBm
LoRa/FSK/OOK

❖ Test Code

```
#include <SPI.h>
#include <LoRa.h>

#define NSS_PIN   10
#define RST_PIN   9
#define DIO0_PIN  2

void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoRa RA-02 Test");

  LoRa.setPins(NSS_PIN, RST_PIN, DIO0_PIN);

  if (!LoRa.begin(433E6)) {   // 433 MHz
    Serial.println("LoRa init failed!");
    while (1);
  }

  Serial.println("LoRa init successful");
}

void loop() {
  // -------- TRANSMIT --------
  Serial.println("Sending: SmartElex LoRa Test");
  LoRa.beginPacket();
  LoRa.print("SmartElex LoRa Test");
  LoRa.endPacket();

  delay(2000);

  // -------- RECEIVE --------
  int packetSize = LoRa.parsePacket();
  if (packetSize) {
    Serial.print("Received: ");

    while (LoRa.available()) {
      Serial.print((char)LoRa.read());
    }

    Serial.print(" | RSSI: ");
    Serial.println(LoRa.packetRssi());
  }

  delay(2000);
}
```

❖ Applications

- Used in long-range IoT and sensor network applications.
- Suitable for industrial telemetry and remote monitoring.
- Enables wireless communication in agriculture and environmental sensing.